GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:	0.16609623	48	6.950e−05	9.618e−02	NDUFB8:103 ECSIT:239 NDUFA9:398 TMEM126B:437 NDUFB6:488 TMEM126A:533
RNA Splicing, Via Transesterification Re	−0.11519753	106	4.318e−05	9.618e−02	RBMX2:3 PRPF40A:145 SNRNP27:179 SRRM1:186 XAB2:311 SRSF2:371
Mitochondrial Respiratory Chain Complex	0.16609623	48	6.950e−05	9.618e−02	NDUFB8:103 ECSIT:239 NDUFA9:398 TMEM126B:437 NDUFB6:488 TMEM126A:533
Monocarboxylic Acid Catabolic Process (G	0.25658984	20	7.142e−05	9.618e−02	PPARD:982 AGXT:1105 NUDT19:1116 LPIN1:1509 CYP26C1:1514 LPIN3:2539
Odontogenesis (GO:0042476)	−0.17410795	36	3.026e−04	3.260e−01	BCOR:377 MSX1:448 COL1A1:624 WNT10A:652 LAMB1:781 INHBA:854
Positive Regulation Of Amyloid−Beta Clea	−0.41474124	6	4.346e−04	3.902e−01	LRP1:258 IL4:324 TREM2:391 ROCK1:1359 APOE:2624 LRPAP1:2699
Alpha−Amino Acid Catabolic Process (GO:1	0.18166929	29	7.127e−04	4.799e−01	HIBCH:471 HMGCL:612 ALDH4A1:676 TAT:1748 KMO:1858 GOT2:1948
mRNA Processing (GO:0006397)	−0.08562310	132	7.030e−04	4.799e−01	RBMX2:3 PRPF40A:145 SNRNP27:179 SRRM1:186 XAB2:311 SRSF2:371
Visual Perception (GO:0007601)	0.10239206	90	8.027e−04	4.804e−01	TRPM1:12 CRYBB2:27 CRYAA:43 CRYBB1:139 CRYZ:148 VAX2:154
mRNA Splicing, Via Spliceosome (GO:00003	−0.07884419	135	1.605e−03	5.086e−01	RBMX2:3 PRPF40A:145 SNRNP27:179 SRRM1:186 XAB2:311 SRSF2:371
Mitochondrial Respiratory Chain Complex	0.10460349	77	1.529e−03	5.086e−01	NDUFB8:103 ECSIT:239 NDUFA9:398 TMEM126B:437 NDUFB6:488 TMEM126A:533
Negative Regulation Of Intrinsic Apoptot	−0.25495752	14	9.582e−04	5.086e−01	KDM1A:542 TAF9:636 ARMC10:703 TAF9B:820 PTTG1IP:1004 ING2:1175
Negative Regulation Of Intrinsic Apoptot	−0.28883582	10	1.564e−03	5.086e−01	KDM1A:542 TAF9:636 TAF9B:820 ING2:1175 ZNF385A:2967 SIRT1:3526
Negative Regulation Of Microtubule Polym	−0.18116135	27	1.126e−03	5.086e−01	TPX2:267 TRIM54:323 MAP2:446 MAPRE1:449 KATNB1:465 NAV3:474
Negative Regulation Of Protein Catabolic	−0.13616862	45	1.588e−03	5.086e−01	AZIN2:295 HMGCR:533 HSP90AB1:578 PTPN3:600 AGAP2:622 SERPINB12:908
Regulation Of Intrinsic Apoptotic Signal	−0.28883582	10	1.564e−03	5.086e−01	KDM1A:542 TAF9:636 TAF9B:820 ING2:1175 ZNF385A:2967 SIRT1:3526
Sister Chromatid Segregation (GO:0000819	0.16818738	30	1.438e−03	5.086e−01	SMC4:14 KIF18B:115 SPAG5:253 ZWINT:272 KIFC1:333 ESPL1:1008
Connective Tissue Development (GO:006144	−0.14893398	37	1.729e−03	5.115e−01	HYAL2:78 VPS13B:434 SOX6:640 CHI3L1:648 SPTLC2:765 SNORC:776
Mitochondrial Electron Transport, NADH T	0.16202346	31	1.804e−03	5.115e−01	NDUFB8:103 NDUFA9:398 NDUFV3:403 NDUFB6:488 NDUFA6:700 NDUFB5:702
Positive Regulation Of Type I Interferon	−0.12094645	55	1.937e−03	5.217e−01	FLOT1:259 DHX58:345 POLR3B:382 TLR7:452 RAB2B:476 TRIM56:508
Regulation Of Amyloid−Beta Clearance (GO	−0.22875995	15	2.162e−03	5.546e−01	LRP1:258 IL4:324 TREM2:391 HMGCR:533 ROCK1:1359 IFNG:1538
7−Methylguanosine Cap Hypermethylation (	−0.35037507	4	1.523e−02	6.121e−01	SNRPG:1287 SNRPD3:1452 SNRPD1:1583 TGS1:4673 NA NA
G Protein−Coupled Glutamate Receptor Sig	0.22107720	11	1.113e−02	6.121e−01	TRPM1:12 GRM6:271 HOMER2:550 GRM8:630 GRM2:672 GRM7:1174
Golgi To Plasma Membrane Transport (GO:0	0.10979990	41	1.505e−02	6.121e−01	GOLGA4:40 MACF1:863 OSBPL5:1519 ANK3:1653 PREPL:1667 NSF:1809
RNA Capping (GO:0036260)	−0.35037507	4	1.523e−02	6.121e−01	SNRPG:1287 SNRPD3:1452 SNRPD1:1583 TGS1:4673 NA NA
RNA Processing (GO:0006396)	−0.07231650	122	5.915e−03	6.121e−01	SRRM1:186 DIS3:253 PPP1R9B:354 SRSF2:371 CELF2:503 METTL3:632
RNA−mediated Gene Silencing (GO:0031047)	0.17432845	22	4.659e−03	6.121e−01	MOV10L1:80 PIWIL1:130 FKBP6:213 TNRC6B:470 DICER1:1003 MOV10:1534
UV Protection (GO:0009650)	−0.30524021	7	5.164e−03	6.121e−01	XPA:436 SCARA3:1591 FEN1:2007 MFAP4:3551 SDF4:3667 MC1R:4515
Aerobic Respiration (GO:0009060)	0.11215933	53	4.772e−03	6.121e−01	NDUFB8:103 NDUFA9:398 NDUFV3:403 NDUFB6:488 NDUFA6:700 NDUFB5:702
Ammonium Homeostasis (GO:0097272)	0.32057971	5	1.304e−02	6.121e−01	RHAG:308 RHCG:429 RHD:905 RHCE:1757 RHBG:10080 NA
Ammonium Transmembrane Transport (GO:007	0.21812367	10	1.693e−02	6.121e−01	RHAG:308 RHCG:429 RHD:905 RHCE:1757 SLC12A4:3372 SLC12A2:4964
Antigen Receptor−Mediated Signaling Path	−0.07441540	97	1.146e−02	6.121e−01	NCKAP1L:93 NFKBID:239 NCOR2:269 CD3G:358 RBCK1:390 THY1:413
Astrocyte Activation (GO:0048143)	−0.24132759	12	3.800e−03	6.121e−01	C5AR1:241 LRP1:258 TREM2:391 IFNG:1538 TNF:1567 PSEN1:1778
Axonemal Central Apparatus Assembly (GO:	−0.36772379	5	4.404e−03	6.121e−01	DNAJB13:79.5 RSPH9:1056 HYDIN:1109 SPEF1:1833 SPAG17:5903 NA
Cardiac Muscle Cell Myoblast Differentia	0.33181586	5	1.018e−02	6.121e−01	RBPJ:300 TBXT:654 NRG1:1444 SRF:5118 NOTCH1:5158 NA
Cardiac Muscle Hypertrophy (GO:0003300)	0.29256544	6	1.307e−02	6.121e−01	CSRP3:968 HDAC4:1247 RYR2:1406 TCAP:2584 HTR2B:3103 OCRL:9399
Cartilage Development (GO:0051216)	−0.10455077	47	1.322e−02	6.121e−01	HYAL2:78 SOX6:640 CHI3L1:648 SNORC:776 SCX:780 COL11A2:808
Cellular Lipid Catabolic Process (GO:004	0.16295438	23	6.842e−03	6.121e−01	PLD1:123 ANGPTL3:127 PPARD:982 NUDT19:1116 LPIN1:1509 SMPDL3B:2200
Cellular Response To Amino Acid Starvati	−0.12774648	37	7.201e−03	6.121e−01	CDKN1A:332 DEPDC5:717 RRAGD:832 MTOR:1026 WDR24:1128 DAPL1:2086
Cellular Response To Bacterial Lipopepti	0.33241262	5	1.005e−02	6.121e−01	TLR10:529 TLR1:1248 CD36:2773 TLR2:3942 TLR6:4096 NA

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
IBRAHIM_NRF1_UP	−0.06673889	357	1.597e−05	5.614e−02	NUP62:12 MAP4:25 EDEM3:63 DNAJC10:65 MRGBP:102 KIF3B:133
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	0.18123310	47	1.731e−05	5.614e−02	NDUFB8:103 ECSIT:239 NDUFV3:403 TMEM126B:437 NDUFB6:488 NDUFA6:700
REACTOME_COMPLEX_I_BIOGENESIS	0.16484010	48	7.829e−05	1.270e−01	NDUFB8:103 ECSIT:239 NDUFA9:398 NDUFV3:403 TMEM126B:437 NDUFB6:488
WP_INTERFERON_TYPE_I_SIGNALING_PATHWAYS	−0.17646211	42	7.628e−05	1.270e−01	JAK1:198 MAP2K3:592 IRF9:620 EIF4E:749 GAB2:762 EIF4EBP1:968
KIM_TIAL1_TARGETS	−0.22889504	24	1.039e−04	1.349e−01	SMARCE1:87 PLSCR3:250 ADSL:343 BDH1:889 ADGRG1:983 FGL1:1020
IBRAHIM_NRF2_UP	−0.04836247	450	4.667e−04	1.791e−01	NCOA7:24 STK40:33 EDEM3:63 DNAJC10:65 PPT1:128 KIF3B:133
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	0.27650944	15	2.091e−04	1.791e−01	EP400:87 P2RX2:349 SFSWAP:681 POLE:1101 GOLGA3:1239 GALNT9:1515
CHEN_LVAD_SUPPORT_OF_FAILING_HEART_UP	−0.10679159	87	5.812e−04	1.791e−01	HYAL2:78 ACTA1:91 SLC38A2:240 FOSL2:279 CDKN1A:332 LAMB1:781
BONCI_TARGETS_OF_MIR15A_AND_MIR16_1	−0.11283135	78	5.759e−04	1.791e−01	PCDHAC2:7 HAS2:363 CDCA4:565 PTPN3:600 BTRC:634 KIF21A:912
LU_EZH2_TARGETS_UP	−0.06442789	231	7.645e−04	1.791e−01	SCAND1:50 HYAL2:78 OASL:110 HDGFL2:231 FAM25A:245 FLOT1:259
FISCHER_DIRECT_P53_TARGETS_META_ANALYSIS	−0.06085034	256	8.283e−04	1.791e−01	RFK:21 RRAD:104 CNOT6:118 TANC1:210 LRP1:258 APAF1:320
FLORIO_NEOCORTEX_BASAL_RADIAL_GLIA_DN	0.07801637	172	4.258e−04	1.791e−01	SMC4:14 DLGAP5:74 KIF18B:115 SPAG5:253 ZWINT:272 MIS18BP1:315
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	−0.04225577	581	5.442e−04	1.791e−01	TUBAL3:79.5 SNAP23:81 PSMD8:131 KIF3B:133 ASB13:192 HERC3:206
REACTOME_HOST_INTERACTIONS_OF_HIV_FACTOR	−0.10056151	93	8.119e−04	1.791e−01	NUP62:12 PSMD8:131 PSMB10:453 BTRC:634 ATP6V1H:668 PSMD3:822
REACTOME_INNATE_IMMUNE_SYSTEM	−0.03593705	770	7.711e−04	1.791e−01	GLIPR1:27 NHLRC3:36 SLC2A3:37 AHSG:43 ASAH1:72 SNAP23:81
REACTOME_METABOLISM_OF_POLYAMINES	−0.16072400	42	3.146e−04	1.791e−01	PSMD8:131 AZIN2:295 PSMB10:453 PSMD3:822 PSMB11:951 ODC1:1221
REACTOME_REGULATION_OF_MRNA_STABILITY_BY	−0.13255877	61	3.450e−04	1.791e−01	PSMD8:131 SET:252 DIS3:253 PSMB10:453 PSMD3:822 EXOSC7:869
REACTOME_TP53_REGULATES_TRANSCRIPTION_OF	−0.15491842	40	7.005e−04	1.791e−01	CNOT6:118 CNOT11:170 CDKN1A:332 PLK2:361 CNOT3:599 RBL1:744
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	−0.14241656	54	2.962e−04	1.791e−01	TUBAL3:79.5 PSMD8:131 CDKN1A:332 MAPRE1:449 PSMB10:453 HSP90AB1:578
REACTOME_CELLULAR_RESPONSE_TO_CHEMICAL_S	−0.08475338	151	3.318e−04	1.791e−01	PRDX2:20 PSMD8:131 NCOR2:269 SCO2:305 CDKN1A:332 PSMB10:453
WP_TYPE_II_INTERFERON_SIGNALING	−0.19085371	30	2.976e−04	1.791e−01	JAK1:198 IRF9:620 CIITA:899 IFIT2:949 H4C14:955 CYBB:956
FULCHER_INFLAMMATORY_RESPONSE_LECTIN_VS_	−0.04629433	493	4.642e−04	1.791e−01	GLIPR1:27 KCNK6:75 SUN2:86 CNPY2:97 TAPT1:130 ACSL3:152
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	−0.03057504	1149	5.660e−04	1.791e−01	GMCL1:49 DLD:59 DRAP1:64 SMARCA5:66 SNAP23:81 CD47:82
UDAYAKUMAR_MED1_TARGETS_DN	−0.07365359	202	3.167e−04	1.791e−01	CDKN1C:23 SLC2A3:37 ELF3:85 GABARAPL1:159 HOXC13:200 UPF1:205
SATO_SILENCED_BY_METHYLATION_IN_PANCREAT	−0.05528054	326	6.284e−04	1.791e−01	CDKN1C:23 CCL5:123 HOXC13:200 HERC3:206 KLK11:214 NCOR2:269
NAKAYAMA_SOFT_TISSUE_TUMORS_PCA1_UP	−0.12322436	62	7.967e−04	1.791e−01	FAP:69 NCKAP1L:93 CCL5:123 LUM:375 HLA−DPA1:694 HLA−DRA:868
BORCZUK_MALIGNANT_MESOTHELIOMA_UP	−0.06324231	239	7.817e−04	1.791e−01	PPT1:128 PSMD8:131 RPS15A:167 TPX2:267 UBQLN2:294 COPA:352
SMIRNOV_RESPONSE_TO_IR_6HR_UP	−0.08850775	135	3.913e−04	1.791e−01	RRAD:104 ANO3:190 APAF1:320 CDKN1A:332 PLK2:361 TOB1:420
BIOCARTA_PLATELETAPP_PATHWAY	−0.26034396	14	7.443e−04	1.791e−01	F9:509 F2:512 PLAU:619 F11:723 PLAT:734 COL4A4:770
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS	−0.09461020	106	7.737e−04	1.791e−01	CBLC:124 HERC3:206 SYVN1:325 UBE2J1:381 FBXO4:468 BTRC:634
GRAESSMANN_RESPONSE_TO_MC_AND_SERUM_DEPR	−0.07249715	174	9.912e−04	1.814e−01	CPEB3:45 ZNF503:67 LGALS3BP:77 ACTB:146 SEPTIN6:207 NMNAT1:225
REACTOME_PD_1_SIGNALING	−0.25413225	14	9.939e−04	1.814e−01	CD3G:358 CD274:519.5 HLA−DPA1:694 CD3E:823 HLA−DRA:868 LCK:1021
REACTOME_HOMOLOGY_DIRECTED_REPAIR	0.09250409	107	9.581e−04	1.814e−01	RNF168:191 CLSPN:230 HERC2:280 BRCA1:395 WRN:418 POLH:454
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	−0.03992379	589	1.007e−03	1.814e−01	RPS26:3 NUP62:12 PRDX2:20 TUBAL3:79.5 CA9:109 PSMD8:131
REACTOME_CELLULAR_RESPONSE_TO_STARVATION	−0.10211568	89	8.780e−04	1.814e−01	RPS26:3 RPL7:157 RPS15A:167 ATP6V1H:668 DEPDC5:717 TCIRG1:779
BECKER_TAMOXIFEN_RESISTANCE_UP	−0.15433338	38	9.973e−04	1.814e−01	LGALS3BP:77 ACTA1:91 EBAG9:191 IRF9:620 BTG3:1254 TAGLN:1258
HERNANDEZ_MITOTIC_ARREST_BY_DOCETAXEL_2_	−0.13379826	46	1.698e−03	2.173e−01	CD47:82 HHEX:155 LUM:375 PLAU:619 PLAT:734 KLF9:835
NIKOLSKY_BREAST_CANCER_8P12_P11_AMPLICON	0.13167007	50	1.283e−03	2.173e−01	NSD3:76 IDO2:77 RAB11FIP1:99 ADAM9:144 DKK4:211 TACC1:226
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	−0.03989573	555	1.404e−03	2.173e−01	NUP62:12 OASL:110 CCL5:123 PSMD8:131 TNFRSF9:166 JAK1:198
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	−0.07918372	132	1.706e−03	2.173e−01	RPS26:3 PSMD8:131 RPL7:157 RPS15A:167 PSMB10:453 CAP2:514

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cataract, Pulverulent	0.41370118	9	1.727e−05	7.021e−02	CRYBB2:27 CRYBB1:139 HSF4:150 GJA8:248 BFSP2:656 VIM:1134
Glioma	−0.03203929	1646	2.864e−05	7.021e−02	TYMP:11 NUP62:12 PDGFB:14 PRDX2:20 CDKN1C:23 GLIPR1:27
Nuclear cataract	0.21571935	32	2.428e−05	7.021e−02	CRYBB2:27 CRYAA:43 CRYBB1:139 HSF4:150 CRYBB3:227 GJA8:248
Nuclear non−senile cataract	0.23396434	29	1.306e−05	7.021e−02	CRYBB2:27 CRYAA:43 CRYBB1:139 CRYBB3:227 GJA8:248 MIP:415
CATARACT, AUTOSOMAL DOMINANT	0.37431299	10	4.158e−05	8.154e−02	CRYAA2:11 CRYBB2:27 CRYAA:43 CRYBB1:139 GJA8:248 MIP:415
Medullary carcinoma of thyroid	−0.08957154	173	5.087e−05	8.313e−02	TYMP:11 MET:181 TNFSF10:339 ALCAM:356 PROM1:359 RXFP2:457
Cataract, Central Saccular, With Sutural	0.46084046	6	9.259e−05	8.911e−02	CRYBB2:27 GJA8:248 MIP:415 BFSP2:656 CRYBA1:662 CRYGS:1486
Malaria, Falciparum	−0.13584876	68	1.091e−04	8.911e−02	MST1:62 CCL5:123 IL4:324 CHI3L1:648 LMLN:666 CD27:690
Carcinoma of lung	−0.02908051	1859	6.487e−05	8.911e−02	CFTR:3 FKBP10:9 TYMP:11 PDGFB:14 DTYMK:16 NABP2:19
Defective tooth enamel	0.18722671	36	1.021e−04	8.911e−02	GTF2IRD1:287 CENPJ:382 PEX6:635 PLK4:866 PEX1:1000 PLEC:1084
Dystrophic tooth enamel	0.18722671	36	1.021e−04	8.911e−02	GTF2IRD1:287 CENPJ:382 PEX6:635 PLK4:866 PEX1:1000 PLEC:1084
Primary malignant neoplasm of lung	−0.02991116	1703	7.425e−05	8.911e−02	CFTR:3 FKBP10:9 TYMP:11 DTYMK:16 NABP2:19 PRDX2:20
Abnormality of dental enamel	0.17120474	40	1.807e−04	1.266e−01	GTF2IRD1:287 CENPJ:382 PEX6:635 PLK4:866 PEX1:1000 PLEC:1084
Malignant neoplasm of lung	−0.02756778	1828	1.695e−04	1.266e−01	CFTR:3 FKBP10:9 TYMP:11 DTYMK:16 NABP2:19 PRDX2:20
Non−Small Cell Lung Carcinoma	−0.02821184	1606	2.662e−04	1.740e−01	CFTR:3 TYMP:11 NUP62:12 PRDX2:20 CDKN1C:23 SLC2A3:37
CATARACT, COPPOCK−LIKE	0.34576715	9	3.284e−04	1.789e−01	CRYBB2:27 CRYBB1:139 GJA8:248 BFSP2:656 VIM:1134 GJA3:2824
Neoplasm Metastasis	−0.02218974	2915	3.020e−04	1.789e−01	CFTR:3 C1GALT1:8 TYMP:11 NUP62:12 PDGFB:14 PRDX2:20
Sacral dimples	0.22156494	22	3.224e−04	1.789e−01	GTF2IRD1:287 NSD2:455 NELFA:556 CLIP2:1187 LETM1:1192 ASXL1:1331
Liver carcinoma	−0.02268547	2600	3.873e−04	1.999e−01	CFTR:3 C1GALT1:8 TYMP:11 NUP62:12 PDGFB:14 NABP2:19
Coxiella burnetii Infection	−0.40729521	6	5.503e−04	2.627e−01	FLOT1:259 HMGCR:533 LDLR:537 LSS:1189 TNF:1567 DHCR24:4223
First myocardial infarction	−0.31499377	10	5.626e−04	2.627e−01	F2:512 LDLR:537 CPB2:569 PLAT:734 PON2:797 IL1RN:1342
Stomach Diseases	−0.21389792	21	6.928e−04	3.088e−01	S100A8:415 HGF:490 GCG:914 AQP3:979 ODC1:1221 SMOX:1489
Dermatitis, Irritant	−0.23286065	17	8.892e−04	3.459e−01	CXCR3:229 IL4:324 CD80:418 HLA−DPA1:694 TRIM10:1286 TNF:1567
Idiopathic pulmonary arterial hypertensi	−0.06472618	226	8.413e−04	3.459e−01	CD47:82 CA9:109 CCL5:123 POU5F1:136 PAM16:213 APLNR:330
Malignant tumor of colon	−0.02609539	1542	9.171e−04	3.459e−01	CFTR:3 C1GALT1:8 TYMP:11 DTYMK:16 PRDX2:20 CDKN1C:23
Nausea	−0.14619005	43	9.173e−04	3.459e−01	TYMP:11 UGT1A1:608 NT5C:752 DHDDS:886 HTR3A:1249 RET:1283
Anemia, hereditary spherocytic hemolytic	0.41384232	5	1.352e−03	3.850e−01	EPB42:332 ANK1:468 SPTA1:756 SLC4A1:2195 SPTB:2665 NA
Night blindness, congenital stationary	0.19207837	23	1.433e−03	3.850e−01	PDE6B:2 TRPM1:12 USP11:113 GRM6:271 CACNA2D4:360 ABCA4:641
Cataract microcornea syndrome	0.30463778	9	1.553e−03	3.850e−01	CRYBB2:27 CRYAA:43 CRYBB1:139 GJA8:248 CRYBA4:1140 CRYGD:3210
Congenital total cataract	0.24276855	15	1.134e−03	3.850e−01	CRYBB2:27 CRYAA:43 HSF4:150 GJA8:248 GCNT2:371 MIP:415
HIV Infections	−0.03908041	591	1.331e−03	3.850e−01	PDGFB:14 DTYMK:16 HNF1A:35 FAP:69 CD47:82 CCL5:123
Lip and Oral Cavity Carcinoma	−0.04832152	372	1.482e−03	3.850e−01	CDKN1C:23 LGALS3BP:77 CA9:109 CCL5:123 CTHRC1:315 IL4:324
Neuroblastoma	−0.02670575	1330	1.462e−03	3.850e−01	TYMP:11 NUP62:12 DTYMK:16 PRDX2:20 CDKN1C:23 NCOA7:24
Osteosarcoma of bone	−0.03609205	721	1.134e−03	3.850e−01	PDGFB:14 DTYMK:16 PRDX2:20 CD47:82 CA9:109 CDKN3:117
Pelvic kidney	0.28815276	10	1.604e−03	3.850e−01	GTF2IRD1:287 CLIP2:1187 FGFR1:1540 RFC2:1551 GTF2I:1666 STRA6:2945
Pierre Robin Syndrome	−0.19795562	22	1.312e−03	3.850e−01	DGCR2:162 COL11A2:808 CD6:919 DGCR6:1202 AMER1:1220 MYMK:1265
Prostate carcinoma	−0.02115122	2344	1.454e−03	3.850e−01	CFTR:3 NUP62:12 PDGFB:14 DTYMK:16 PRDX2:20 CYP7B1:22
Retinal pigment epithelial atrophy	0.24704877	14	1.374e−03	3.850e−01	CERKL:334 PRPH2:606 ABCA4:641 CNGB3:836 ELOVL4:2098 RS1:2371
Secondary malignant neoplasm of lymph no	−0.03105033	932	1.610e−03	3.850e−01	TYMP:11 PRDX2:20 CDKN1C:23 GATA5:42 DLD:59 MST1:62
Spherocytosis	0.41384232	5	1.352e−03	3.850e−01	EPB42:332 ANK1:468 SPTA1:756 SLC4A1:2195 SPTB:2665 NA

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0002169 no abnormal phenotype	−0.02925535	1083	3.134e−03	4.825e−01	CFTR:3 PCDHAC2:7 FAP:69 SUN2:86 RABGAP1L:98 CBLC:124
MP0005551 abnormal eye electrophysiolog	0.07914660	136	1.613e−03	4.825e−01	PDE6B:2 CPLX3:4 TRPM1:12 GABRR1:50 CDHR1:55 NAGLU:96
MP0003252 abnormal bile duct	0.18139323	21	4.063e−03	4.825e−01	PKHD1:412 ABCB11:651 ABCB4:879 PKD2:1518 NR1H4:1591 ABCG5:1681
MP0002873 normal phenotype	−0.02925240	1088	3.087e−03	4.825e−01	CFTR:3 PCDHAC2:7 FAP:69 SUN2:86 RABGAP1L:98 CBLC:124
MP0002452 abnormal antigen presenting	−0.03227453	539	1.419e−02	6.126e−01	CFTR:3 RFK:21 GPR183:89 NCKAP1L:93 SASH3:100 TNFRSF9:166
MP0001286 abnormal eye development	−0.06344121	140	1.036e−02	6.126e−01	CDKN1C:23 MAB21L2:70 VAX1:141 HHEX:155 PITX3:178 DISP1:255
MP0001853 heart inflammation	−0.14565495	25	1.186e−02	6.126e−01	NCKAP1L:93 RELA:693 GJA1:985 IRAK1:1144 TREX1:1763 CHD2:2122
MP0001819 abnormal immune cell	−0.02750516	822	1.247e−02	6.126e−01	CFTR:3 RFK:21 CD47:82 GPR183:89 NCKAP1L:93 SASH3:100
MP0003221 abnormal cardiomyocyte apopto	−0.12869091	33	1.070e−02	6.126e−01	TRIM54:323 DNAJA3:644 BACH1:794 NOX4:1379 SLC8A1:1515 MAPK9:1977
MP0001666 abnormal nutrient absorption	0.11299319	40	1.368e−02	6.126e−01	DRD2:105 SOAT2:231 ABCG8:736 ABCB4:879 SLC23A1:1289 ABCG5:1681
MP0002420 abnormal adaptive immunity	−0.02737995	818	1.306e−02	6.126e−01	CFTR:3 RFK:21 CD47:82 GPR183:89 NCKAP1L:93 SASH3:100
MP0005085 abnormal gallbladder physiolo	0.14276802	24	1.566e−02	6.197e−01	GPBAR1:277 ABCB11:651 ABCG8:736 ABCB4:879 CYP27A1:959 F2RL1:1285
MP0002723 abnormal immune serum	−0.03009096	595	1.697e−02	6.200e−01	LGALS3BP:77 GPR183:89 NCKAP1L:93 SASH3:100 CCL5:123 SCARB1:132
MP0000001 mammalian phenotype	−0.01819359	2791	2.628e−02	6.912e−01	CFTR:3 PCDHAC2:7 TYMP:11 PDGFB:14 CDKN1C:23 EXT2:26
MP0001929 abnormal gametogenesis	0.03251584	410	2.884e−02	6.912e−01	PHYH:21 PHC2:66 MOV10L1:80 RLIM:128 PIWIL1:130 FOXN1:145
MP0002932 abnormal joint morphology	−0.06601638	98	2.499e−02	6.912e−01	HYAL2:78 SYVN1:325 HAS2:363 COL1A1:624 LRP6:698 PLAT:734
MP0002272 abnormal nervous system	0.05049697	175	2.301e−02	6.912e−01	PHYH:21 GABRR1:50 DRD2:105 KCNC2:254 KCNQ3:323 KCNQ2:408
MP0000432 abnormal head morphology	−0.03971238	285	2.410e−02	6.912e−01	CFTR:3 WNT9B:76 HYAL2:78 VAX1:141 SP3:184 FGF6:242
MP0000653 abnormal sex gland	0.03210069	457	2.342e−02	6.912e−01	MOV10L1:80 DRD2:105 PIWIL1:130 FOXN1:145 FANCM:192 FKBP6:213
MP0010094 abnormal chromosome stability	0.08391860	57	2.910e−02	6.912e−01	FOXN1:145 FANCG:224 BRCA1:395 ASZ1:728 BARD1:838 BUB1B:953
MP0008438 abnormal cutaneous collagen	0.16376341	14	3.405e−02	7.033e−01	COL3A1:167 COL5A2:1251 PPIB:1934 FBN1:3222 LOX:3465 COL5A1:3793
MP0003436 decreased susceptibility to	−0.09968916	38	3.396e−02	7.033e−01	ALOX5AP:139 CLEC5A:221 SYVN1:325 CDKN1A:332 PLAU:619 DUSP2:786
MP0002118 abnormal lipid homeostasis	−0.03291086	380	3.263e−02	7.033e−01	CFTR:3 CYP7B1:22 HNF1A:35 AHSG:43 GPD2:57 ASAH1:72
MP0002080 prenatal lethality	−0.01985391	1206	3.730e−02	7.269e−01	C1GALT1:8 PDGFB:14 RFK:21 CDKN1C:23 EXT2:26 HNF1A:35
MP0005379 endocrine/exocrine gland phen	−0.14533493	17	3.826e−02	7.269e−01	CFTR:3 CDKN1C:23 PRLR:1184 MC5R:2215 PAX7:2327 E2F1:2615
MP0002128 abnormal blood circulation	−0.03147324	378	4.151e−02	7.584e−01	C1GALT1:8 PDGFB:14 MET:181 LRP1:258 TRIM54:323 CDKN1A:332
MP0001697 abnormal embryo size	−0.03145925	364	4.524e−02	7.960e−01	DLD:59 KIF3B:133 MET:181 HS3ST1:224 DISP1:255 SMAD1:349
MP0002210 abnormal sex determination	0.03328756	302	5.206e−02	8.153e−01	MOV10L1:80 DRD2:105 PIWIL1:130 FOXN1:145 FANCM:192 FKBP6:213
MP0002138 abnormal hepatobiliary system	0.19592040	8	5.512e−02	8.153e−01	PKHD1:412 ABCB4:879 SOCS2:1746 ARG1:2060 MAP3K14:4724 ACVRL1:6713
MP0005253 abnormal eye physiology	0.05845015	89	5.836e−02	8.153e−01	TRPM1:12 NR2E3:247 GJA8:248 EGFLAM:264 MIP:415 PRPH2:606
MP0008875 abnormal xenobiotic pharmacok	0.13886536	16	5.474e−02	8.153e−01	CYP1A1:484 CYP1A2:530 MTF1:1052 AS3MT:2214 AHR:2987 COMT:3602
MP0002106 abnormal muscle physiology	−0.04767613	147	4.852e−02	8.153e−01	ACTA1:91 CACNA2D1:212 FGF6:242 CDKN1A:332 RICTOR:546 FBXO32:594
MP0002653 abnormal ependyma morphology	−0.18307679	9	5.734e−02	8.153e−01	VAX1:141 E2F5:913 DLG5:1086 HYDIN:1109 NUMB:2753 DLL3:2838
MP0006054 spinal hemorrhage	−0.19917910	8	5.120e−02	8.153e−01	C1GALT1:8 COL1A1:624 PSEN1:1778 C1GALT1C1:3804 SPHK1:4007 FLI1:5333
MP0000609 abnormal liver physiology	−0.02926623	360	6.384e−02	8.193e−01	HNF1A:35 GPD2:57 HYAL2:78 NCKAP1L:93 SCARB1:132 PITX3:178
MP0000604 amyloidosis	−0.08962424	33	7.549e−02	8.193e−01	NCKAP1L:93 LDLR:537 MSR1:986 DLST:1599 PSEN1:1778 MAPT:2090
MP0001765 abnormal ion homeostasis	0.05263621	99	7.248e−02	8.193e−01	TRPM6:3 ATP2A2:229 CLDN16:426 RHCG:429 NEDD4L:505 PTH1R:562
MP0001845 abnormal inflammatory respons	−0.02480243	475	7.478e−02	8.193e−01	CFTR:3 C1GALT1:8 LGALS3BP:77 NCKAP1L:93 SASH3:100 ALOX5AP:139
MP0008877 abnormal DNA methylation	0.13715026	14	7.589e−02	8.193e−01	MOV10L1:80 UHRF1:275 ASZ1:728 ARID4A:995 RNF4:3758 MAEL:4067
MP0004808 abnormal hematopoietic stem	0.06936060	58	6.890e−02	8.193e−01	EP400:87 MPL:186 CSF3R:442 ITGA2B:561 ARID4A:995 MECOM:1121

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
eye.development	−0.104774103	57	6.361e−03	3.435e−01	ELF3:85 VAX1:141 PITX3:178 SMAD1:349 LUM:375 MSX1:448
amygdala	−0.143304291	8	1.606e−01	8.187e−01	PCDH15:56 SLC38A8:2596 NEUROD2:2846 PTPRZ1:4325 TNR:5009 BCAN:5833
atrium.auricle	0.026920169	89	3.821e−01	8.187e−01	TNNI3K:46 XIRP1:826 LMOD3:878 CSRP3:968 BMP10:983 MYOM2:1012
blood	0.025186711	266	1.630e−01	8.187e−01	JAK3:8 BPI:194 RHAG:308 TREML2:329 EPB42:332 DHX34:343
breast	−0.095224137	14	2.176e−01	8.187e−01	GSC:2183 CIDEC:3278.5 ADIPOQ:3671 LEP:4152 FABP4:4389 ABCB5:4722
C1.spinal.cord	−0.021171415	171	3.437e−01	8.187e−01	CTNNA3:203 ARHGAP22:293 PLD4:340 TREM2:391 TMEM235:696 ACSBG1:721
cerebellar.hemisphere	0.017615701	459	2.067e−01	8.187e−01	PHTF1:78 ADCY1:137 VAX2:154 NBEA:163 JMJD7:172 CADM3:190
cortex.kidney	0.034900064	110	2.085e−01	8.187e−01	SLC5A12:36 CLCNKB:116 CUBN:367 PKHD1:412 CLDN16:426 DCDC2:560
esophagogastric.junction	−0.090057109	8	3.779e−01	8.187e−01	HOXA4:853 ADCY5:3019 NKX6−1:3649 CHRM3:3925 BARX1:4380 COL4A5:6510
esophagus.mucosa	0.028850843	213	1.511e−01	8.187e−01	ADGRF1:23 SOX15:89 RAB11FIP1:99 DSG3:102 FOXN1:145 CRB3:170
esophagus.muscularis.mucosa	−0.095586325	26	9.198e−02	8.187e−01	EPHA7:257 FAM83D:268 PALLD:739 S1PR3:1029 KCNMB1:1967 SYNPO2:2064
fallopian.tube	0.035630554	48	3.942e−01	8.187e−01	ADAMTS3:24 CCDC17:58 ADGRG4:244 GMNC:296 GREB1L:1492 GASK1A:1972
greater.omentum	−0.054279941	32	2.886e−01	8.187e−01	PRRT4:10 GFPT2:187 MEST:378 TCF15:1074 FAM89A:1593 RBP7:2780
hippocampus.proper	−0.100501204	9	2.966e−01	8.187e−01	NTS:2270 GRP:2551 NEUROD2:2846 NEUROG3:4107 FEZF2:5452 OLIG2:5905
hypothalamus	−0.057042286	58	1.340e−01	8.187e−01	PCDH15:56 GSX1:834 SYNGR3:1006 NGB:1017 HAP1:1079 PMCH:1262
liver	0.025383849	322	1.235e−01	8.187e−01	ACY1:26 THPO:33 SERPINA7:48 ACOX2:52 ADGRG7:56 IDO2:77
ovary	0.027500598	98	3.492e−01	8.187e−01	CPZ:216 PLIN5:477 TSHZ3:565 CCDC68:589 NLGN4X:626 SYTL4:675
pancreas	0.043123956	93	1.525e−01	8.187e−01	CPB1:5 TMED6:125 ERP27:159 CELA3B:208 KCNK16:380 CTRC:414
pituitary.gland	0.017640475	202	3.923e−01	8.187e−01	OTOG:68 IGSF1:86 VEPH1:101 DRD2:105 REC8:386 RIMBP2:404
prostate	0.046287758	37	3.307e−01	8.187e−01	NR2E3:247 P2RX2:349 SCUBE2:460 CPAMD8:520 SHROOM1:613 CREB3L4:1529
putamen	−0.088199577	14	2.535e−01	8.187e−01	ANO3:190 SLC18A3:1457 GRM3:1776 OPALIN:2884 FAM237A:3160 GBX2:4190
testis	−0.012644560	1339	1.507e−01	8.187e−01	DNAH2:3 C11orf42:47 BTG4:53 DIAPH3:58 DNAI2:74 DNAJB13:79.5
thyroid	0.029958411	147	2.133e−01	8.187e−01	TLE6:9 CLCNKA:57 PRTG:61 IDO2:77 DGKI:153 CRYBB3:227
transverse.colon	−0.022846735	144	3.474e−01	8.187e−01	HNF1A:35 TUBAL3:79.5 ATP10B:122 R3HDML:140 PYY:447 LEFTY1:496
uterus	0.043245483	53	2.773e−01	8.187e−01	CLEC4F:263 TEX15:352 ADAMTS19:731 CHRDL2:899 HOXD4:1269 MRGPRF:1503
vagina	0.043551814	89	1.574e−01	8.187e−01	DSG3:102 FOXN1:145 PGLYRP4:250 SLC18A2:310 RHCG:429 DSC2:602
Brodmann.area.9	0.011644241	162	6.120e−01	9.228e−01	MAS1:122 TMEM132D:141 NETO1:210 KCNC2:254 GPR26:260 KCNQ3:323
caudate.nucleus	−0.041313260	26	4.664e−01	9.228e−01	VAX1:141 ANO3:190 SLC35D3:926 GRM3:1776 SLC1A2:3005 FAM237A:3160
EBV.lymphocyte	−0.007665488	652	5.190e−01	9.228e−01	GPR183:89 NCKAP1L:93 SASH3:100 OASL:110 ANKRD44:147 HELB:164
ectocervix	−0.035713447	25	5.370e−01	9.228e−01	KCNK6:75 KRT34:811 PRLR:1184 MATN2:1347 GDF7:2315 TRHDE:4098
leg.skin	0.010699497	209	5.978e−01	9.228e−01	TRPM1:12 IL17RE:28 CDHR1:55 NKPD1:92 DSG3:102 FOXN1:145
lung	−0.017146748	139	4.883e−01	9.228e−01	CLEC5A:221 GPRC5D:441 ESM1:458 HGF:490 NDNF:774 ICAM1:974
minor.salivary.gland	−0.018944650	69	5.876e−01	9.228e−01	PROM1:359 CRACR2A:548 SERINC2:590 BPIFB6:683 ATP6V1B1:783 ELF5:938
substantia.nigra	−0.058919106	11	4.988e−01	9.228e−01	C10orf105:844 RET:1283 SLC6A3:2558 SPX:3208 TH:7018 KLHL1:7822
suprapubic.skin	−0.012702783	181	5.592e−01	9.228e−01	KRTDAP:15 CBLC:124 FAM25A:245 ANXA8L1:463 GRHL2:471 GAL3ST4:558
transformed.skin.fibroblast	−0.009478626	243	6.152e−01	9.228e−01	FKBP10:9 CYP7B1:22 FAP:69 KIRREL1:108 SLC38A2:240 LRP1:258
nucleus.accumbens	−0.013217643	93	6.610e−01	9.647e−01	VAX1:141 ANO3:190 DRD1:504 FAT3:538 NECAB2:910 COCH:947
left.ventricle	0.013340241	67	7.066e−01	9.784e−01	TNNI3K:46 XIRP1:826 LMOD3:878 CSRP3:968 MYOM2:1012 SMCO1:1127
tibial.artery	0.011013676	101	7.035e−01	9.784e−01	SUSD5:31 MN1:185 CRLF1:259 RERGL:306 SDK1:316 WFDC1:317
aorta	0.007035557	151	7.671e−01	9.863e−01	SUSD5:31 GABRR1:50 KCNK15:200 CNTN4:342 GAS6:366 HMCN1:389